

Nirmit Aggarwal

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Education

Maharaja Surajmal Institute of Technology (MSIT)
Bachelor of Technology in Information Technology

2025–2029

Technical Skills

Languages: JavaScript, TypeScript, Rust, Python, SQL, Bash

Backend Engineering: Node.js, Express.js, REST APIs, JWT, OAuth, WebSockets, Redis, API Design, Multer

Frontend: React, Vite, Tailwind CSS, Zustand

Databases: MongoDB, MySQL, Redis

DevOps & Infrastructure: Docker, GitHub Actions, CI/CD, Linux

Developer Tools: Git, GitHub, Postman, npm, Bun, VS Code

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design, Distributed Systems

Software Engineering Experience

Technical Lead – IEEE MSIT

Apr 2026–Present

- Led a team of six developers to build the Student Toolkit platform for IEEE MSIT.
- Planned project architecture, coordinated GitHub workflow, assigned features, and reviewed contributions.
- Implemented GitHub OAuth, JWT authentication, PDF/Image utilities, and the initial application foundation.

Software Engineering Intern – MegaBuilders Program

Jun 2026–Jul 2026

- Contributed to backend development, debugging, feature implementation, and collaborative software engineering workflows.

Student Developer – GDG MSIT, Geek Room, Microsoft Student Chapter

Aug 2025–Aug 2026

- Participated in hackathons, technical workshops, collaborative software projects, and developer communities.

Selected Projects

Volunteer Cloud

GitHub(nirmitAggarwal/volunteer_cloud)

Rust, Tokio, Actix Web, SQLite, Python

- Architected a distributed volunteer computing platform inspired by BOINC/Folding@home.
- Built asynchronous worker orchestration with heartbeat monitoring, automatic recovery, and benchmark-aware scheduling.
- Designed a secure plugin execution pipeline using Ed25519 signatures and SHA-256 verification.
- Developed both a **Python** prototype and a high-performance **Rust** implementation with monitoring dashboard.

Student Toolkit

GitHub(IEEE-MSIT/student_toolkit)

React, Express.js, MongoDB, GitHub OAuth, JWT

- Led a six-member IEEE MSIT engineering team building a production-ready student productivity platform.
- Implemented GitHub OAuth authentication, JWT session management, PDF and Image utilities.
- Established the initial frontend structure and collaborated on backend architecture.

Open Source

rust-ui/ui

- Contributed a Leptos troubleshooting guide improving Windows and Git Bash developer onboarding.
- Reported implementation issues and collaborated through GitHub pull requests and issue discussions.